

BARRA, United KingdomWMO: **030350**Lat: **57.033N**Lon: **7.450W**Elev: **10**StdP: **14.69**Time Zone: **0.00 (EUW)**Period: **97-07**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	33.5	35.1	25.8	19.8	37.5	27.6	21.6	38.2	50.0	42.4	46.9	43.2	11.1	10	0.835

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.4	65.9	61.2	63.6	59.5	62.0	58.3	62.3	64.8	60.7	62.5	59.4	61.1	12.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.1	80.4	63.7	59.7	76.5	61.8	58.5	73.2	60.4	28.0	65.0	26.9	62.7	26.0	61.1	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.0	36.8	33.4	DB	28.6	71.0	5.8	3.2	24.4	73.3	21.0	75.2	17.8	77.0	13.5	79.4	(4)
(5)			WB	26.6	64.6	5.2	2.2	22.8	66.2	19.8	67.5	16.8	68.7	13.0	70.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	44.4	43.7	44.7	47.0	50.6	54.1	56.8	58.2	55.9	51.3	47.8	45.0
	DBStd	6.10	3.43	4.02	3.71	3.91	3.18	3.09	2.88	2.74	2.66	3.52	3.57	4.10
	HDD50	926	173	178	167	103	31	3	1	0	1	29	82	158
	HDD65	5480	638	597	630	541	446	328	253	213	273	425	516	620
	CDD50	922	1	2	1	12	50	126	213	253	177	69	15	3
	CDD65	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.6	20.4	20.5	17.7	15.8	13.7	13.9	12.2	12.8	15.6	18.2	19.4	19.4
Precipitation	PrecAvg	51.3	6.2	4.5	4.1	3.1	2.6	2.8	3.2	4.0	4.1	5.4	5.5	5.5
	PrecMax	67.6	10.3	10.4	8.5	5.0	6.4	5.3	5.6	10.5	8.1	9.4	9.1	9.5
	PrecMin	37.8	2.6	1.9	0.7	1.3	0.3	0.9	1.3	1.7	1.1	1.8	3.5	1.1
	PrecStd	6.8	1.9	2.1	1.6	0.9	1.4	1.1	1.0	1.8	1.7	2.2	1.3	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	53.1	53.4	59.9	64.7	67.1	69.3	69.8	63.8	59.3	56.4	54.5
		MCWB	51.5	51.5	50.7	52.4	57.8	60.7	62.8	64.4	59.3	56.8	54.5	52.1
	2%	DB	51.8	51.9	52.0	55.7	59.8	62.4	64.9	66.5	62.1	58.1	55.0	53.0
		MCWB	50.6	50.3	49.7	50.6	54.4	57.4	60.8	62.3	58.3	55.7	52.9	51.3
	5%	DB	50.7	50.7	50.9	53.7	57.1	60.3	62.5	63.9	60.9	57.2	53.8	51.9
		MCWB	49.3	49.0	48.7	50.1	53.2	56.2	59.2	60.3	57.9	54.8	51.5	50.3
	10%	DB	49.5	49.4	49.8	52.1	55.3	58.7	61.0	62.3	59.8	56.2	52.8	50.8
		MCWB	47.8	47.5	47.7	49.3	52.0	55.2	57.8	59.0	57.0	53.8	50.6	49.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	52.4	51.9	53.4	58.0	60.8	64.0	65.1	61.2	58.0	55.1	53.3
		MCDB	52.5	52.8	52.5	56.7	62.8	66.0	68.1	69.0	62.4	58.5	55.9	53.8
	2%	WB	51.0	51.0	50.7	51.9	55.4	58.4	61.9	62.9	59.7	56.8	53.9	51.9
		MCDB	51.5	51.7	51.5	54.1	58.2	60.9	64.2	65.4	60.9	57.6	54.7	52.7
	5%	WB	49.7	49.5	49.4	50.8	53.8	57.2	60.1	61.3	58.7	55.6	52.3	50.8
		MCDB	50.4	50.4	50.3	53.0	56.2	59.4	61.8	63.3	60.2	56.6	53.2	51.7
	10%	WB	48.2	47.8	48.3	49.7	52.6	56.0	58.8	59.9	57.6	54.4	51.2	49.4
		MCDB	49.3	49.2	49.3	51.8	54.7	58.0	60.4	61.6	59.2	55.9	52.4	50.6
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.3	6.8	7.6	8.2	7.7	7.5	7.4	6.6	5.7	5.8	6.1
		MCDBR	6.7	5.9	7.0	8.9	10.2	9.9	9.2	9.4	7.5	5.9	5.7	5.9
		MCWBR	7.4	6.2	6.0	6.1	7.0	6.8	5.9	6.4	5.7	5.7	6.8	7.0
	5% WB	MCDBR	6.3	5.8	6.3	7.8	9.2	8.6	8.4	8.8	6.4	5.8	5.8	6.1
		MCWBR	7.2	6.4	6.2	6.0	6.8	6.8	6.0	6.7	5.6	6.6	7.3	7.5
Clear-Sky Solar Irradiance	taub	0.355	0.361	0.377	0.387	0.390	0.377	0.380	0.370	0.373	0.355	0.340	0.334	
	taud	2.132	2.238	2.285	2.316	2.366	2.445	2.449	2.483	2.434	2.363	2.209	2.116	
	Ebn at Noon	150	212	244	262	269	274	271	266	246	214	157	123	
	Edh at Noon	19	25	31	35	35	33	32	29	27	22	17	15	
All-Sky Solar Radiation	RadAvg	124	323	698	1212	1589	1622	1488	1193	785	409	164	84	
	RadStd	13	31	51	82	119	161	102	79	42	41	15	10	